Operating Manual

Industriemotoren - Industrial Engines 2009 ►

Otto engine (petrol, LPG/CNG)

Edition 05.2012

Enginecode	CEEA	CKYA	BEF	CBS	BMF	CKPA			

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Please acknowledge, therefore, that we must reserve the right to amend the product at any time in terms of the configurations equipment and technology.

The data provided concerning scope of delivery, appearance, performance, dimensions, weights, fuel consumtion, standards and functions are all correct at the time of going to print.

Some of equipment described might not yet be available in a particular (information can be provided by your local Volkswagen dealership), and some equipment may not be available in certan countries.

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Operating Manual

1 Preface

1.1 Welcome!

Thank you for purchasing a Volkswagen industrial engine.

Worldwide, several hundred thousand Volkswagen industrial engines have proven their reliability in stationary and mobile applications.

Volkswagen industrial engines have the following benefits:

- Compact design and low weight mean that they can be installed in a wide variety of applications.
- Long service life and low fuel consumption mean they are very economical.

Of the products available on the market, you chose an industrial engine that is advanced in every respect.

Your industrial engine has been developed for minimum impact on the environment.

One of largest and most efficient service organisations is available to service your Volkswagen industrial engine:

Volkswagen Service

Volkswagen Service is cost-efficient and works to factory specifications. It ensures that your industrial engine remains free of faults. In addition, Volkswagen Service offers a comprehensive warranty package and service offers.

Volkswagen Service and the Volkswagen importers will be happy to provide information on warranty terms and claims specific to individual countries .

Volkswagen AG

1.2 About this manual

This manual contains information about the engine. The information provided must be observed without fail. The respective manufacturer is responsible for details about the machine or the vehicle in which the engine is to be installed. Observe the associated documentation and instructions.

All of the details referred to in this operating manual are based on the information available at the time of going to print. Due to constant developments, there may be discrepancies between the engine and the details in this manual. The discrepancies in the details, illustrations or descriptions shall not form the basis of any legal claim.

The “illustrations” may deviate slightly from the parts/equipment installed in your vehicle/machine. These should therefore be regarded as a general guide only.

Some sections of this booklet do not apply to all vehicles/machines. In this case, the area of validity should be indicated at the start of the section “by way of the engine type of the engine code (e.g. CJDA)”.

Correct handling, maintenance and care of the Volkswagen drive system serves as a means of value retention and is also one of the conditions required for warranty claims. For further information about warranties, see ???. ⇒Page 30

Note

We recommend that a suitable specialist workshop is entrusted to perform the service measures. Relevant work instructions and training courses can be obtained from your Volkswagen service partner responsible for drive systems.

WARNING

For all work measures on the engine or in the engine compartment, observe the general safety information.
⇒Page 3

1.3 Signs and symbols

WARNING

Texts with this symbol contain information about your safety and how risks can be avoided that could otherwise cause fatal or life-threatening injuries.

The “Caution” symbol requires special attention. Make sure that these texts are “always” read.

Caution

Texts with this symbol contain information about how damage to the engine can be avoided.

A “Beware” symbol draws your attention to the fact that if certain information is not heeded, damage to the engine could ensue (e.g.: Ensure that the ignition is switched off before disconnecting the battery, otherwise the engine control unit could be damaged).

Note

- *Texts with this symbol contain additional, useful information.*
- *A “Note” symbol refers to special additional details about the repair measure and information associated with it.*

2 General safety instructions

2.1 Safety notes for working on engine

Note

- *Before performing measures on the engine that could well be covered by the comprehensive warranty, always seek advice from a qualified specialist workshop.*
- *Incorrect parts or improperly performed work can cause serious damage to the engine and its systems. They can also necessitate expensive repair measures to rectify the damage and the warranty cover could be impaired as a result.*
- *Before performing repair work on a vehicle/machine, ensure that all necessary tools and replacement parts are ready to hand. Read all instructions and descriptions carefully and heed all „CAUTION“ texts. Observe the repair instructions without omitting any individual steps. Always use the tools and equipment described in the manual that comply with Volkswagen specifications. Auxiliary tools, incorrect parts and improper work procedures lead to unprofessional repairs.*
- *Part numbers stated in this manual serve purely as a reference. Always check these part numbers with the parts department of a qualified specialist workshop to ensure they reflect the latest information.*
- *Exercise due care and attention to protect the environment. Before draining engine oil, for example, ensure that it can be collected and disposed of in the correct manner. Under no circumstances should service fluids be disposed of or allowed to seep in the ground, in the sewage/drainage system, in a river, stream, pond or lake. If there are any queries, contact the person responsible for environmental protection in your company or local or district council office.*

⚠ WARNING

Exercise particular caution when working on the vehicle/ machine. Adopt a common sense approach and follow the general regulations and guidelines for health and safety in the workplace. Never take any risks that could cause an injury.

If you do not have the specialist skills required to carry out the work described in these instructions, we strongly recommend that you entrust a qualified specialist workshop to do it. This recommendation also applies in cases where individuals and workshops are not equipped with the necessary tools and equipment.

Observe the workshop safety precautions. Always wear protective eyewear when working with tools and equipment or acid. In addition, always wear protective eyewear, protective gloves and suitable protective clothing when handling hazardous materials.

Never work on a vehicle/machine if you do not feel up to the task. If you are suffering from fatigue or stress, you increase the risk of injuring yourself or others. Likewise, some medication or pharmaceutical products can affect your concentration and impair your ability to react in good time.

Make sure that your working area has sufficient and appropriate lighting. Use a portable safety lamp when working inside or underneath the vehicle/machine. Also make sure that the bulb is sufficiently shielded. The hot filament of a damaged bulb could start a fire and lead to burns in the event of fuel or oil spillage.

⚠ WARNING

Observe the inherent risk of burning, injuries, accident and fire when working on the engine or in the engine compartment.

Never start the engine of a vehicle/machine in an area with limited space or in an enclosed room with poor and inadequate ventilation. Engine emissions contain carbon monoxide (CO), a colourless, odourless gas that can kill.

Never leave the vehicle/machine unattended if the engine is running. An unattended vehicle/machine with running engine is a hazard and can cause injury.

The engine compartment of any vehicle/machine poses a potential risk to safety. Before carrying out inspection work in the engine compartment, stop the engine, withdraw the ignition key from the ignition lock and allow the engine to cool down. Hot engine components can burn skin on contact.

Do not open the coolant expansion tank while the engine is hot. The cooling system is pressurised.

WARNING

If work needs to be carried out on the electrical system or fuel system of the vehicle/machine, always disconnect the negative cable of the vehicle/machine battery. Under no circumstances is smoking permitted in the vicinity of heaters, heating equipment or naked flames. Fluids in the engine compartment could catch fire. Always have a working fire extinguisher on hand.

Fluids such as fuel, oil, brake fluid, etc. must always be collected in containers that are suitable and have the appropriate marking. To avoid the risk of poisoning, do not use food containers or drinks cans. A person unaware of the contents could be tempted to drink from them.

To reduce the risk of fire, always keep flammable liquids away from any source of fire. If fluids are spilt they should be cleaned up immediately. Cleaning cloths should likewise be stored and disposed of in the correct manner.

Avoid contact with service fluids. If service fluids come into contact with the eyes, rinse the eyes immediately with clear water and notify a doctor without delay. In cases such as these, keep the original container and show it to the doctor.

WARNING

If working with tools in the engine compartment or in or around the area while the engine is running, always tie up long hair. Never wear a tie, scarf, neck chain, loose clothing or similar items. These could get caught in the moving parts of an engine, including the poly-V-belt, alternator and radiator fan. This could result in serious injury.

Do not wear any jewellery when working with machinery, tools or in the engine compartment. This could cause short circuit or electric shock. Likewise, jewellery could get caught up in rotating parts or could be crushed by heavy items and thereby cause serious injury.

3 Technical data

Engine number/engine code ⇒ Page 6.

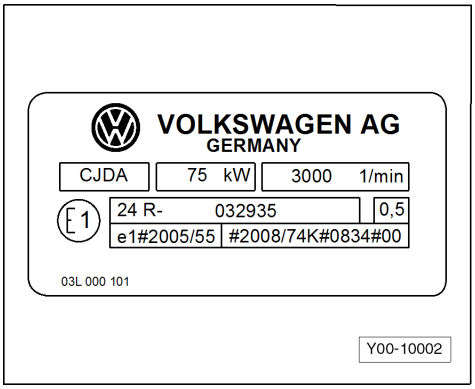
Engine data ⇒ Page 6.

In the event of any queries, claims/complaints and when ordering replacement parts, always indicate the engine designation, engine code and engine number. This helps to avoid any mistakes when dealing with the request.

> ***WARNING***
>
> ***Before reading off the details, switch off the engine and allow it to cool down.***
>
> ***For all work measures on the engine or in the engine compartment, observe the safety information.*** ⇒Page 3

3.1 Engine number/engine code

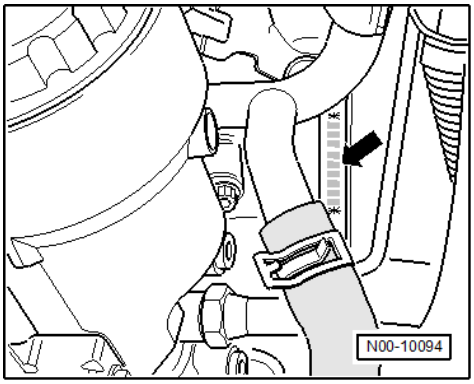
The engine code can be found on the factory plate, which is located on the side of the toothed belt guard or on the timing case.



The engine number („engine code“ and „serial number“) can be found on the joint -arrow-.

In addition, an adhesive label can be found on the toothed belt guard indicating the „engine code“ and „serial number“ .

The engine number consists of up to nine digits (alphanumeric). The first part (maximum 3 letters) represents the „engine code“, the second part (six digit) represents the „serial number“. If more than 999999 engines are produced with the same code, the first of the six digits is replaced by a letter.



3.2 Engine data

Petrol engines

Engine codes		CEEA	CKYA
Production		12.08 ►	01.11 ►
Engine designation		G 1.2 345	G 1.2 345
Engine type		Petrol engine	Petrol engine
Capacity	l	1.2	1.2
Output	kW at rpm	45/4200-5800	45/4200-5800
Torque	Nm at rpm	108/3250	108/3250
Bore	∅ mm	76.5	76.5
Stroke	mm	86.9	86.9
Compression ratio		10.3	10.3
Fuel	at least	95 unleaded, in exceptional cases 91 RON.	95 unleaded, in exceptional cases 91 RON.

Engine codes		CEEA	CKYA
Firing order		1-2-3	1-2-3
Emissions thresholds in accordance with		-	-
Upper idling speed	rpm	3500/5950	3500/5950
Lower idling speed	rpm	1000 +/- 50	1000 +/- 50

Spark ignition engines, prepared for gas operation

Engine codes		BEF	CBS	BMF	CKPA
Production		►	►	►	►
Engine designation		LPG 2.0 - 4XX	CNG 2.0 - 4XX	LPG 3.2 - 655	LPG 3.6 - 658/672
Engine type		Spark ignition engine, prepared for gas operation	Spark ignition engine, prepared for gas operation	Spark ignition engine, prepared for gas operation	Spark ignition engine, prepared for gas operation
Capacity	l	2.0	2.0	3.6	3.6
Output	kW at rpm	Performance range 40-44 kW , depending on customer application	Performance range 38-41 kW, depending on customer application	55/2500, depending on customer application	58/1875 72/2500, depending on customer application
Torque	Nm at rpm	143/2200	137/2200	225/1400	287/1800
Bore	∅ mm	82.5	82.5	84	89
Stroke	mm	92.8	92.8	95.9	96.4
Compression ratio		10.5	10.5	11.25	9.7
Fuel	at least	LPG complying with EN 589, RON 95	CNG, MZ 70	LPG complying with EN 589, RON 95	LPG complying with EN 589, RON 95
Firing order		1-3-4-2	1-3-4-2	1-5-3-6-2-4	1-5-3-6-2-4
Emissions thresholds in accordance with		no registration	no registration	no registration	no registration
Upper idling speed	rpm	4000	4000	2800	3200
Lower idling speed	rpm	800	800	1000	1000

4 Operation

4.1 Warning lamps and indicator lamps

The warning and indicator lamps draw the driver's attention to warnings, malfunctions or certain functions. Some warning and

indicator lamps light up when the ignition is switched on and should go out once the engine is running.

> ⚠ ***WARNING***
>
> ***The control elements of the engine are the responsibility of the machine/engine manufacturer. These also include the warning and indicator lamps. Make sure you read the information provided by the machine/vehicle manufacturer.***

Not every vehicle/machine manufacturer uses the same warning and indicator lamps and this also applies to the graphics or typographical conventions. The actual warning and indicator lamps displayed, their meaning and the remedial action required must be gleaned from the relevant documentation of the vehicle/machine manufacturer.

Note

- *The exact details about the meaning of the transmitted warning and indicator signals from the engine control unit and the remedial action necessary are made available to machine/vehicle manufacturers in the documentation.*
- *Failure to observe illuminated indicator lamps and text messages could lead to engine damage.*

Signals are commonly associated with the following warning and indicator lamps

- Engine oil warning lamp
- Battery warning lamp
- Engine warning lamp

> ⚠ ***WARNING***
>
> ***Failure to observe the warning lamps and text messages could lead to breakdown, accidents and serious injuries.***
>
> - ***Never ignore any warning lamps or text messages.***
> - ***Before opening the flap to the engine compartment, switch off the engine and allow it to cool down sufficiently.***
> - ***The engine compartment is a hazardous area and can cause serious injuries.***

5 Starting and stopping the engine

Introduction

This chapter contains information on the following subjects:

⇒„Starting the engine“ on page 10

⇒„Switching off the engine“ on page 10

⇒„Engine malfunction“ on page 10

- Before starting, check the engine oil level, coolant level and fuel level. Replenish as necessary.
- At low temperatures and on vehicles or machines connected to the engine via a manual clutch, the clutch pedal must be fully depressed when starting so that the starter motor only turns the engine.
- As soon as the engine starts, release the key or starter button immediately - the starter motor must not be turned by the engine.
- If the ignition lock is equipped with a starter interlock feature and the starting procedure needs to be repeated or the engine has stalled, restarting is only possible once the key has been turned back fully. The starter interlock feature prevents the starter motor from engaging when the engine is running, as this could cause damage.
- Avoid high engine speeds and maximum acceleration until the engine has reached its operating temperature.
- When starting from cold, the engine may be a little noisy for the first few seconds until engine oil pressure has built up in the hydraulic valve lifters. This is quite normal and is no cause for concern.

⚠ WARNING

The risk of serious injury when the engine is running or when the engine is started can be reduced.

- ***Never start or run the engine in unventilated or enclosed rooms. The exhaust fumes contain carbon monoxide, for example, an odourless and colourless poisonous gas. Carbon monoxide can cause loss of consciousness and death.***
- ***Never exit the control area of the vehicle or the mobile machine while the engine is running. The vehicle or mobile machine could move suddenly, particularly if a gear is engaged, and cause accidents and serious injuries.***
- ***Never use starting fluid. Starting fluid could explode and cause the engine to turn over suddenly.***

⚠ WARNING

The components of the exhaust system become very hot. This could lead to fires and serious injuries.

- ***Never park the vehicle where parts of the exhaust system can come into contact with any inflammable materials underneath the vehicle or machine, e.g. branches/ twigs, leaves, dry grass, spilt fuel and so on.***
- ***Never apply additional underseal or anti-corrosive treatment to the exhaust pipes, catalytic converter or heat shields.***

5.1 Starting the engine

Note

First read and observe the preliminary information and safety warnings ⇒ Page 8.

> ***WARNING***
>
> ***Never exit the control area of the vehicle or the mobile machine while the engine is running. The vehicle or mobile machine could move suddenly, particularly if a gear is engaged, and cause accidents and serious injuries.***

Note

- *When the engine is cold, avoid high engine speeds, full acceleration and high engine load.*
- *Do not allow the engine to warm up when the vehicle is stationary.*
- *Components with a high power consumption should be switched off temporarily when the engine is started.*

Note

There is a risk of poisoning if the engine is run in enclosed spaces. The engine exhaust gas contains carbon monoxide, an odourless and colourless gas. Inhaling this gas is hazardous to health.

5.2 Switching off the engine

Note

First read and observe the preliminary information and safety warnings ⇒ Page 8.

If the engine has been run for a considerable time under high load, do not switch it off immediately. Instead, allow it to run at idling speed for approx. 2 minutes to allow it to cool down.

Note

If the vehicle/machine has been operated under heavy load for a long period, the engine could overheat when it is switched off. In order to avoid damage to the engine, allow the engine to run in neutral for approximately two minutes before switching it off.

5.3 Engine malfunction

Note

First read and observe the preliminary information and safety warnings ⇒ Page 8.

If, during operation, a malfunction occurs in the engine management system, this can be seen by the malfunction indicator

lamp flashing. The engine should be checked in a specialist workshop without delay.

In addition, a message could be shown in the display of the instrument cluster with relevant information or instructions to perform certain tasks.

Note

To avoid damage to the engine, always observe the indicator lamps and corresponding information.

6 Operating instructions

Preparations for work in the engine compartment

6.1 Introduction

The engine compartment is a hazardous area. Never carry out any work on the engine or in the engine compartment if you are not familiar with the necessary procedures and the general safety requirements and if the workshop equipment, service fluids and tools are not suitable for the task. If there is any doubt, have the work carried out in a specialist workshop. Serious injuries can be caused if work is carried out incorrectly.

⚠ WARNING

The engine compartment is a hazardous area and can cause serious injuries.

- ***The utmost care and attention must be paid when carrying out any work and the general safety rules must be observed. Never take any risks.***
- ***Never undertake any work on the engine or in the engine compartment without the proper skills and experience. If there is any doubt, have the work carried out in a specialist workshop. Serious injuries can result if work is not carried out in the correct manner.***
- ***Never open or close the engine compartment flap if steam or coolant is escaping. Hot steam or coolant can cause severe scalding. Always wait until the escaping steam or coolant has subsided and can no longer been seen or heard.***
- ***Always allow the engine to cool down before opening the engine compartment flap.***
- ***Hot components from the engine or exhaust system can burn the skin on contact.***
- ***Once the engine has cooled down, observe the following before opening the engine compartment flap.***

- Secure the vehicle or machine to prevent it from moving. - Apply the handbrake if necessary.

- Remove the key from the ignition lock.

- Always keep children away from the engine compartment and never leave the vehicle unattended.

- ***When refilling, do not spill any service fluids on engine components or on the exhaust system. Spilt service fluids could start a fire. In some circumstances, the ethylene glycol in the engine coolant could catch fire.***
- ***The engine cooling system is pressurised when the engine is hot. Never open the cap of the coolant expansion tank when the engine is hot. Coolant could spray out and cause serious burns and injuries.***

- Turn the cap anticlockwise slowly and cautiously while exerting light pressure downwards on the cap.

- Always protect the face, hands and arms from hot coolant or steam using a large, thick cloth.

- ***When refilling, do not spill any service fluids on engine components or on the exhaust system. Spilt service fluids could start a fire.***

⚠ ***WARNING***

High voltage in the electrical system can cause electric shocks, burns, severe and fatal injury.

- ***Never cause a short circuit in the electrical system. The battery could explode.***
- ***To avoid the risk of electric shock and severe injury, observe the following while the engine is running and when it is being started:***

- Never touch electrical leads/cables or wires of the ignition system.

- Never touch the power cables and connections of the gas discharge lamps. There are rotating components in the engine compartment that can cause serious injury.

- ***Never reach into areas of these components or the radiator fan. Contact with the fan blades could result in serious injury. The fan is temperature-controlled and could start automatically, even when the engine has been switched off and the key has been removed from the ignition lock.***
- ***If any work has to be performed when the engine is started or while the engine is running, there is an additional, potentially fatal safety risk from rotating parts (e.g. drive belts, alternator, radiator fan) and from the high voltage ignition system. Always exercise extreme caution.***

- Always ensure that no body parts, jewellery, ties, loose items of clothing or long hair can be caught up in rotating engine components. Before starting work, remove any jewellery and ties, tie up long hair and tuck in clothes tightly to prevent them from getting caught in engine parts.

- Always exercise extreme caution when accelerating the engine; never accelerate the engine arbitrarily. The vehicle or machine could start to move if it has not been adequately secured.

- ***Never leave any objects in the engine compartment, such as cleaning cloths or tools. Disregarded items could cause malfunctions, engine damage and fire.***

WARNING

Service fluids and some of the materials in the engine compartment are highly flammable and can cause fires and serious injuries.

- ***Never smoke.***
- ***Never work near naked flames or sparks.***
- ***Never spill fluids onto the engine. These could catch fire on hot engine components and thereby cause injury.***
- ***Observe the following when carrying out any work on the fuel system or the electrical system:***

- Always disconnect the battery.

- Never work in the vicinity of heating systems, coolant heaters or other naked flames.

- ***Always have a fully functional and tested fire extinguisher to hand.***

WARNING

Ignoring the smell of gas in a vehicle / machine or when refuelling could result in serious injuries.

- ***Take the necessary action.***
- ***Exit the danger zone.***
- ***Contact the emergency services if necessary.***

Note

- *When refilling or changing service fluids, ensure that the contents of the containers are correct. Incorrect service fluids can result in serious malfunctions and engine damage. Observe the information provided by the manufacturer of the bottle/container system.*
- *Service fluid leaks from the vehicle or machine are harmful to the environment. For this reason, check the ground under the vehicle / machine regularly. If engine oil or other fluids are detected on the ground, the vehicle / machine should be checked in a specialist workshop. Any spilt service fluids must be disposed of properly.*

6.2 Fuel

Observe the preparations for work in the engine compartment and follow the preliminary instructions on the topic ⇒Page 11.

⇒„Fuel information for gas engines“ on page 15

⇒„Fuel information for petrol engines“ on page 15

6.2.1 Fuel information for gas engines

Fuel information

Engine	BEF, BMF, CKPA	CBS
Fuel	Liquid petroleum gas (LPG) complying with EN 589	Compressed natural gas (CNG)
Octane rating	RON 95	MZ 70
Composition	Propane-butane mixture with maximum butane content of 80 % and no more than 5 % of unsaturated elements.	Liquid petroleum gas must not be mixed with biogas.

The gas pressure may vary depending on the composition of the gas and the temperature. A high percentage of butane could result in a pressure that is insufficient for engine operation below 0 °C.

WARNING

When designing the fuel tank and fuel lines etc., observe the legislative requirements and bottle manufacturer's instructions. If the regulations are not observed, severe injuries could ensue.

Note

- *Industrial propane gas could contain a high percentage of olefin compounds (unsaturated hydrocarbon compounds). If the amount is excessive, the liquid gas / natural gas system and engine could sustain damage. Fuels of poor quality or low octane rating can damage the engine.*
- *The quality of natural gas varies considerably throughout Europe. Mixing liquid gas or biogas poses a risk to safety and can cause damage to the engine.*

6.2.2 Fuel information for petrol engines

Engine code: CEEA, CKYA

The engine must be operated using unleaded petrol with 95 RON (at least 91 RON) complying with EN 228 (EN = „European Norm“) ⇒Page 16.

Petrol variants are categorised according to their octane number, e.g. 91, 95, 98 or 99 RON (RON = „Research Octane Number“). It is permissible for the engine to be run using petrol with a higher octane number than required. However, this has no advantage in terms of fuel consumption and engine power.

WARNING

Never use „engine starter fluid“. These fluids can be very dangerous and cause the engine to run at high revs, posing a risk of injury.

Caution

- ***Use only fuel that complies with EN 228 with the correct octane number. Otherwise, the engine and the fuel system could suffer severe damage. The engine could also lose power or fail.***
- ***If, in an emergency, the tank has to be replenished using petrol with a lower octane number than recommended, drive in the mid engine speed range and avoid high engine loads. High engine speeds and high engine loads can cause serious damage to the engine. Replenish using petrol with the correct octane number as soon as possible.***

6.3 Engine oil

Introduction

The engine oil level should be checked before and after the vehicle/machine is used.

Checking engine oil level ⇒ Page 17

Topping up engine oil⇒ Page 17

- The engine oil must be changed ⇒ Page 25regularly in accordance with the data given in the service schedule.
- The engine oil and filter change should always be carried out in a specialist workshop as special tools and knowledge are required, this also applies to the disposal of used oil. Volkswagen recommends that these tasks are carried out at a Volkswagen dealership.

Note

- *It the engine is constantly operated in regions with high dust levels or in arctic conditions, i.e. temperatures permanently below approx. 20° C, the engine oil should be changed more frequently. If there is any doubt as to whether certain circumstance require more frequent engine oil changes, seek advice from a Volkswagen service centre.*
- *Additives in the engine oil can cause new engine oil to look darker even after a short space of time. This is normal and does not mean that the engine oil should be changed more frequently.*
- *As with all other service fluids, any spilt engine oil can pollute the environment. Spilt service fluids should be collected and disposed of in the correct manner, without harming the environment.*

WARNING

Incorrect handling of engine oil can cause serious burns and injuries.

- ***Always wear protective eyewear when handling engine oil.***
- ***Engine oil is poisonous and must be stored out of reach of children.***
- ***Engine oil must be kept in the closed original container. This also applies to used oil while it is awaiting disposal.***
- ***Never use empty food containers, bottles or other containers to store engine oil as the contents could be consumed by unsuspecting individuals.***
- ***Regular contact with engine oil can damage the skin. Skin that has been in contact with engine oil should be washed thoroughly with soap and water.***
- ***Engine oil becomes extremely hot when the engine is running and can scald skin severely. Always allow the engine to cool down.***

6.3.1 Checking engine oil

Observe the following points:

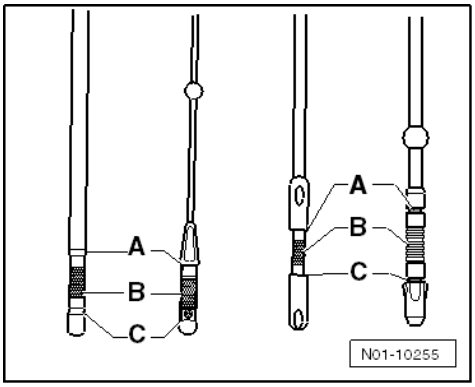
– After stopping the engine, wait at least 3 minutes for the oil to drain back into the sump.

– Pull out the oil dipstick, wipe it on a clean cloth and insert it again, pushing it in as far as it will go.

Note

Observe the disposal regulations.

– Pull the dipstick out again and read off the engine oil level.



Dipstick readings as follows:

A - Engine oil must not be replenished.

B - Engine oil may be replenished. After replenishing, the oil level could then be in segment -A-.

C - Engine oil must be replenished. After filling, it is then adequate for the oil level to be in segment -B- (checkered area).

If the engine oil level is above the -A- marking, there is a risk of damaging the catalytic converter.

– If the engine oil level is below the -C- marking, replenish engine oil up to the -A- marking. Engine oil specification .

6.3.2 Replenishing engine oil

Engine oil specifications and engine oil capacity:

Caution

Only use oils of the correct oil specification that has been approved for the engine.

- ***If engine oils with different oil specifications are used, the engine could be damaged and an accident could ensue.***

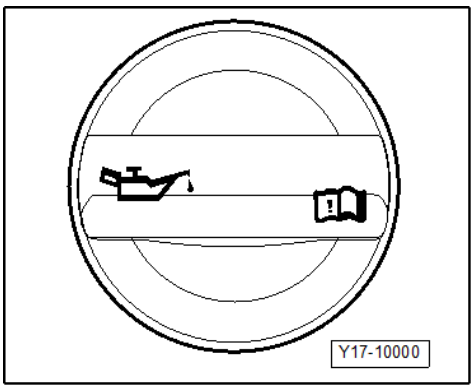
Engine codes	With engine oil filter change	Engine oil specification
BEF	4.5 l	504 00
BMF	5.5 l	504 00
CBS	4.5 l	504 00
CEEA	2.8 l	504 00
CKYA	2.8 l	504 00
CKPA	6.9 l	504 00

General information

Note

Observe the disposal regulations.

Replenishing engine oil



- Unscrew cap of engine filler opening in cylinder head. If there is any doubt regarding the location of the cap, seek expert advice.
- Only replenish with engine oil that has been specifically approved by Volkswagen for this engine and only pour in small quantities at a time.
- To avoid overfilling, wait after pouring in each small quantity so that the engine oil can flow into the engine oil sump and be measured correctly on the engine oil dipstick.
- Check the engine oil level before pouring in the next small quantity of engine oil. Never pour in too much engine oil.
- Once the engine oil level has reached at least segment -B-, push the oil dipstick into the guide tube as far as it will go in order to prevent any engine oil escaping when the engine is running.
- After replenishing, the cap for the engine oil filler opening must be screwed back on again correctly.

– After pouring in the engine oil, wait for at least 3 minutes and then check the oil level.

– Pull out the oil dipstick, wipe it on a clean cloth and insert it again, pushing it in as far as it will go.

– Pull the dipstick out again and read off the engine oil level.

Dipstick readings as follows:

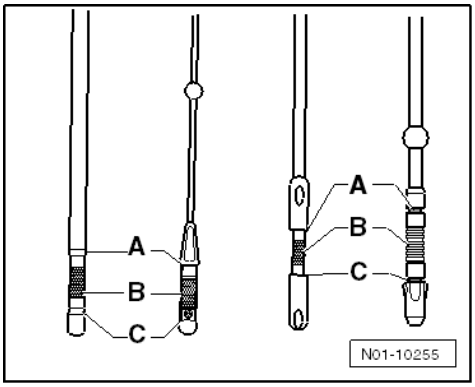
A - Engine oil must not be replenished.

B - Engine oil may be replenished. After replenishing, the oil level could then be in segment -a-.

C - Engine oil must be replenished. After filling, it is then adequate for the oil level to be in segment -B- (checkered area).

If the engine oil level is above the -A- marking, there is a risk of damaging the catalytic converter.

– If the engine oil level is below the -C- marking, replenish engine oil up to the -A- marking.



WARNING

Engine oil can ignite if it comes into contact with hot engine components. It can cause fires, burns and other serious injuries.

- ***Always ensure that the cap of the engine filler opening is securely closed after refilling. This will prevent the engine oil from escaping on to hot engine components when the engine is running.***

Note

- *Do not start the engine if the engine oil level is above segment -A-. Seek expert assistance. The catalytic converter and the engine could otherwise be damaged.*
- *When refilling or changing service fluids, ensure that the contents of the containers are correct. Incorrect service fluids can result in serious malfunctions and engine damage.*
- *The engine oil level must never be above segment -A-. Otherwise, engine oil could be drawn in through the crankcase breather and escape into the atmosphere via the exhaust system.*

6.4 Coolant

Introduction

- The engine coolant should be checked before and after the vehicle/machine is used.
- Checking coolant level, replenishing coolant additive ⇒Page 20

The engine oil must be changed regularly in accordance with the details given in the servicing section⇒Page 25.

WARNING

Please note that the percentage of correct coolant additive must correspond to the lowest outside temperature that the vehicle / machine is expected to be operated in.

Serious injuries can be caused if work is carried out incorrectly.

Insufficient anti-freeze in the engine coolant system can cause the engine to break down and cause serious injuries.

Note

As with all other service fluids, any spilt engine coolant can pollute the environment. Spilt service fluids should be collected and disposed of in the correct manner, without harming the environment.

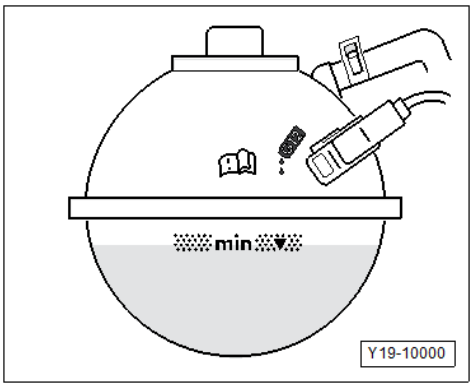
6.4.1 Checking coolant level, replenishing coolant additive

– Check the coolant level in the expansion tank when the engine is cold.

◆ Handover inspection: Coolant level above „MIN marking“ -arrow-.

◆ Inspection service: Coolant level above „MIN marking“ -arrow-.

– If the coolant level is too low, replenish with fluid mixed to the correct ratio.



Note

If fluid loss exceeds the amount expected from normal operation, determine and rectify the cause (repair procedure).

6.4.2 Mixture ratio:

Caution

Only distilled water may be used for mixing. Tap water or well water does not have the requisite properties to safeguard the functionality of the coolant.

Anti-freeze to	Coolant additive G 12+ and G 12++	Water[1)]
-25 °C	approx. 40 %	approx. 60 %
-35 °C	approx. 50 %	approx. 50 %
-40 °C	approx. 60 %	approx. 40 %

1) Only use distilled water.

Note

- *Coolant additives G 12+ and G 12++ prevent frost and corrosion damage, limescale and also raise the boiling point. For this reason, it is essential that the cooling system is filled with radiator anti-freeze and corrosion protection throughout the year.*
- *Particularly in countries with tropical climates, the coolant contributes towards operating safety thanks to the higher boiling point at high engine loads.*
- *The concentration of the coolant must not be reduced by adding distilled water, even during the warm season and in warm countries. The percentage of coolant additive must be at least 40 %.*

7 Electrical system

Introduction

Never carry out any work on the electrical system if you are not familiar with the necessary procedures and the general safety requirements and if the workshop equipment, service fluids and tools are not suitable for the task. If there is any doubt, have the work carried out in a specialist workshop. Serious injuries can be caused if work is carried out incorrectly.

WARNING

Work measures on the battery and electrical system can cause serious chemical burns, fire and electric shocks. Before starting any work, always study the warning notices and safety precautions and heed the information provided by the manufacturer of the electrical system components.

- ***Switch off the ignition and all electrical consumers before carrying out any work on the battery and also disconnect the negative cable from the battery.***
- ***Do not smoke and never work in the vicinity of naked flames or sparks.***
- ***When handling cables and electrical equipment, avoid generating sparks and electrostatic charge.***

Note

Never disconnect batteries if the ignition is switched on or the engine is running and never connect them together as otherwise damage to the electrical system and electronic components could ensue.

8 Engine attachments, replacement parts, repairs and modifications

Introduction

This chapter contains information on the following subjects:

⇒„Engine attachments and replacement parts“ on page 22

⇒„Service fluids and consumables“ on page 23

⇒„Repairs and technical modifications“ on page 24

⇒Page 21

WARNING

Unsuitable replacement parts and engine attachments as well as incorrectly performed work, modifications and repairs can cause damage, accidents and serious injuries.

- ***Volkswagen strongly recommends that only industrial engine approved Volkswagen accessories and Volkswagen Genuine Parts® are used. These parts and accessories have been specially tested by Volkswagen for suitability, reliability and safety.***
- ***Repairs and modifications to the industrial engine should only be carried out in a specialist workshop. Specialist workshops have the necessary tools, diagnostic equipment, repair information and qualified personnel.***
- ***Never fit parts to the industrial engine that are in any way different from the factory-fitted parts.***

8.1 Engine attachments and replacement parts

First read and observe the preliminary information and safety warnings ⇒Page 21.

Volkswagen recommends that advice is sought from a Volkswagen dealership before accessories, replacement parts or service fluids are procured. For example, if the vehicle or machine is to be retrofitted with accessories or if parts have to be replaced. Volkswagen dealerships can provide information about legislative requirements and factory recommendations for engine attachments, replacement parts and service fluids.

Volkswagen recommends that only approved Volkswagen genuine parts® are used. These parts and accessories have been specially tested by Volkswagen for suitability, reliability and safety. A Volkswagen dealership is also qualified to install the parts in the correct manner.

Despite constant market appraisal, Volkswagen cannot assume responsibility for products that have not been approved by Volkswagen in terms of reliability, safety and suitability in the vehicle or machine. Volkswagen can therefore assume no responsibility for these parts, even if they have been approved by an official testing agency or are covered by an official approval certificate.

Procurement of replacement parts

- Replacement parts with a Volkswagen part number can be ordered easily from the special catalogue for industrial engines via the global ETKA system from Volkswagen. These parts are made available within a short space of time.
- Replacement parts for equipment that does originate from Volkswagen (e.g. gas systems) should be ordered from the respective manufacturer.

8.2 Service fluids and consumables

All service fluids and consumables, e.g. toothed belts, tyres, engine coolant, engine oils and also spark plugs and batteries, are subject to constant further development. For this reason, service fluids and consumables should be replaced in a specialist workshop. A Volkswagen dealer is always kept up-to-date regarding innovations.

Coolant⇒Page 19

Engine oil⇒Page 16

WARNING

Unsuitable service fluids and consumables as well as improper use of fluids and consumables could cause accidents, serious injuries, burns or poisoning.

- ***Service fluids must be kept in the closed original container.***
- ***Never store service fluids in empty food containers, bottles or any other non-original containers as the contents could be consumed by unsuspecting individuals.***
- ***Keep children away from all service fluids and consumables.***
- ***Always read and follow the information and warnings on the service fluid packaging.***
- ***When using products that give off harmful fumes, always work outdoors or in a well-ventilated area.***
- ***Never use fuel, turpentine, engine oil, nail varnish remover or other volatile fluids for cleaning and care. These substances are poisonous and highly flammable. They could cause fires and explosions.***

Note

- *Only use suitable service fluids for refilling. Never use the wrong service fluid. Failure to observe this warning can result in serious faults and engine damage.*
- *Accessories and other attachments that impair the inlet of cooling air have a negative effect on the cooling properties of coolant. The engine may overheat at high ambient temperatures and high engine loads.*
- *Leaking service fluids can pollute the environment. Spilt service fluids must be collected in suitable containers and disposed of properly, without harming the environment.*

8.3 Repairs and technical modifications

Note

- *If you do not have the specialist skills required to carry out the work described in these instructions, we strongly recommend that you entrust a qualified specialist workshop to do it. This recommendation also applies in cases where individuals and workshops are not equipped with the necessary tools and equipment.*
- *Before performing repairs on a vehicle/machine that could well be covered by the comprehensive warranty, always seek advice from a qualified specialist workshop.*

9 Troubleshooting

9.1 General notes

One of largest and most efficient service organisations is available to service your Volkswagen industrial engine:

Volkswagen service

Volkswagen Service is cost-efficient and works to factory specifications. It has trained specialists, modern equipment and all the necessary special tools.

Professional advice and fast and effective aid is available here. The following chapter provides information on faults and solutions should the engine does not start, for example. The engine should be taken to a Volkswagen service centre if a repair becomes necessary. There it will receive expert treatment.

9.2 Malfunction: Engine does not start

Observe and follow the preparations for work in the engine compartment ⇒Page 11.

Possible cause	Solution
Fuel tap closed	Open fuel tap
Fuel filter blocked	Renew fuel filter
No fuel in tank	Refuel
Fuel supply problem	Check lines, filter and tank breather. Repair or renew parts as necessary.

The Volkswagen service centre will be happy to help if the remedial action is not successful.

9.3 Malfunction: Idling problems

Observe and follow the preparations for work in the engine compartment ⇒Page 11.

Possible cause	Solution
Fuel hose between fuel pump and fuel filter is loose	Check connections are tight and replace fuel hose if necessary.
Fuel supply problem	Filter contaminated: clean or renew. Fuel return line and injection line leaking, dirty or blocked: repair or replace.

The Volkswagen service centre will be happy to help if the remedial action is not successful.

9.4 Malfunction: Poor performance, maximum speed cannot reached

Observe and follow the preparations for work in the engine compartment ⇒Page 11.

Possible cause	Solution
Air filter/fuel filter contaminated	Clean filter insert, replace as necessary.
Fuel system leaking	Carry out visual check of all lines and repair.
Fuel lines defective	Lines dirty, kinked or restricted, tank breather blocked: repair or replace.
Return line blocked	Blow air through return line from fuel pump to tank with air .

The Volkswagen service centre will be happy to help if the remedial action is not successful.

9.5 Malfunction: Indicator lamp flashing

Observe and follow the preparations for work in the engine compartment ⇒Page 11.

Possible cause	Solution
Preglow indicator lamp flashing	Seek assistance at specialist workshop

The Volkswagen service centre will be happy to help if the remedial action is not successful.

10 Service measures

Note

- *We recommend that service work is only ever carried out in a qualified service workshop. These specialist workshops carry out repairs and maintenance work in the correct manner in accordance with the workshop manual for industrial engines.*
- *Incorrect parts or improperly performed repairs and adjustments can cause serious damage to the vehicle/machine and its systems. They can also necessitate expensive repair measures to rectify the damage and the warranty cover could be impaired as a result.*
- *Observe the safety information* ⇒Page 3

WARNING

The following instructions must be followed without fail.

- ***Use only service fluids and replacement parts that comply with the technical requirements specified by Volkswagen.***
- ***If any replacement parts and accessories are used that have not been supplied by Volkswagen, the specific properties of the machine could be negatively influenced, thereby compromising safety.***
- ***If the points listed above are not observed and damage is caused as a result, any warranty entitlement or claims against Volkswagen will be invalidated.***

Note

⇒„Entries in service record“ on page 26

⇒„Service intervals“ on page 26

⇒„Scope of interval service“ on page 28

10.1 Entries in service record

All service work and repairs that are carried out on the engine must be documented in the separate service record document of the respective engine.

If a component is replaced that is subject to replacement at intervals prescribed by the manufacturer, e.g. the toothed belt, the interval until replacement is due again commences from this point.

- For this reason, it is very important to document any change of components in the service record.
- This also applies to components that were replaced before the usual replacement interval.

Note

- *To decide on the use of „genuine replacement part kits“, consider whether replacement of all of the components included in the genuine replacement part kit is actually necessary.*
- *If more components are going to be replaced than actually necessary, the customer must be made aware of this fact before the repair work is carried out.*

10.2 Service intervals

⇒„Difficult operating conditions“ on page 27

⇒„Extended service intervals“ on page 28

The service intervals based on the operating hours and length of time, which are described in this chapter, were calculated

and determined by Volkswagen. The service intervals can always be reached under normal operating conditions.

Note

- *In the event of a combination of operating hours and time specifications: whichever occurs first applies.*
- *In addition to the interval service, inspection service and interval service inspection, additional maintenance tasks must be performed depending on the operating conditions.*
- *It is also possible to have additional tasks carried out at times other than those scheduled in the service record.*

Note

- *If the way the Volkswagen industrial engine is deployed has not been analysed by way of an engine parameter check, the operating conditions cannot be estimated by Volkswagen.*
- *If there is any doubt regarding the operating conditions, Volkswagen recommends that the maintenance intervals are halved, e.g. 600 to 300 operating hours; 1800 to 900 operating hours; 2400 to 1200 operating hours and so on.*
- *However, there may be operating conditions that necessitate an even greater interval frequency.*

The service interval applies to the engines listed here from the time they are commissioned and delivered.

Service intervals/events:: BEF, BMF, CBS, CEEA, CKPA, CKYA		
All engines		
Service type MKB / PR no. / Remarks	**Service events**	**Intervals**
Engine operation under normal conditions ⇒Page 28	Interval service	every 600 hrs / 1 year , whichever comes first
Engine operation under difficult conditions ⇒Page 27	Interval service	< 600 hours / < 1 year depending on the specific conditions; if in doubt divide the specification in half.
Extended service intervals ⇒Page 28	Interval service	Following written confirmation from your Volkswagen contact person.

10.2.1 Difficult operating conditions

In the event of difficult operating conditions, some work measures must be performed before the scheduled interval service.

This applies in particular to the air filter, toothed belt, fuel filter, and the engine oil.

The following make this necessary:

- Nonobservance of operating instructions ⇒Page 11.
- Fuels of poorer quality.
- Frequent operation in regions with high dust levels.

- Engine operation in regions with extremely low temperatures over a longer period of time.
- Non-stop engine operation.
- Mainly “stop and go” operation.

10.2.2 Extended service intervals

There is an option of extending the service intervals.

If you are interested in extended service intervals, contact your Volkswagen representative for details.

Note

- *By extending the service intervals, maintenance costs can be reduced.*
- *To safeguard the service life of the engine, the machine manufacturer must successfully complete a variety tests.*

10.3 Scope of interval service

⇒„General scope of service“ on page 28

⇒„Additional work“ on page 29

⇒„Scope of service for gas systems“ on page 30

Service tables can be found in the following chapter with Volkswagen-specific maintenance specifications.

10.3.1 General scope of service

Note

- *The engine oil and coolant level should be checked before and after the vehicle/machine is used.*⇒Page 11
- *Unless otherwise agreed, the usual interval service applies in all cases.*
-
- *Here, an engine oil change service must be carried out every 600 hours of operation or after 1 year* ⇒Page 26*.*
- *If there is any doubt regarding the operating conditions* ⇒Page 26 *(service interval chart), Volkswagen recommends that the maintenance intervals are halved, e.g. 600 to 300 operating hours; 1800 to 900 operating hours; 2400 to 1200 operating hours and so on.*

Scope of service

Service type MKB / PR no. / Remarks	Scope of service
◆ Interval service ◆ Applies to all engine codes Every 600 hrs. / 1 year whichever comes first	◆ Self-diagnosis: interrogate fault memories of all systems.
	◆ Change engine oil, replace engine oil filter.
	◆ Visually check engine and components in the engine compartment for leaks and damage.
	◆ Check the coolant level and frost protection; replenish coolant / additives as necessary.
	◆ Test the engine.
	◆ Poly V-belt: check condition; renew as necessary.
	The attachment parts of the engine periphery are subject to a specific interval service. The specifications of the respective manufacturer must be adhered to in all cases. Among these are, e.g.: ◆ Visually inspect radiator, radiator fan and cooling fins for leaks and external contamination. ◆ Check exhaust system for leaks, secure fitting and damage. ◆ Renew air filter ◆ Check hoses/clamps; renew as necessary ◆ Check air lines for leaks; renew as necessary

Note

- *The attachment parts of the engine periphery are subject to an interval service.*
- *Maintenance tasks and interval service operations must be performed in accordance with the manufacturer's specifications.*

10.3.2 Additional work

In addition to the interval service, additional tasks are necessary depending on the operating conditions ⇒Page 27.

It is also possible to have additional tasks carried out at times other than those scheduled in the service record.

Every 600 hrs.

Additional work	Engine code
– Replace spark plugs	CBS BEF CEEA CKYA

Every 1,800 hrs.

Additional work	Engine code
– Replace toothed belt	CBS BEF

Every 3000 hrs.

Additional work	Engine code
– Replace spark plugs	BMF CKPA

Every 6000 hrs.

Additional work	Engine code
– Renewing coolant pump (Volkswagen recommends that the coolant pump is renewed when the toothed belt is changed)	CEEA CKYA
	BMF CKPA

Emissions test

Note

Observe the legislative requirements.

10.3.3 Scope of service for gas systems

WARNING

The legislative requirements for the maintenance of gas systems must be observed when the gas system is maintained. Only properly trained personnel may perform the tasks.

Note

The relevant maintenance intervals for the gas system can be obtained from the manufacturer of the gas system.

11 Warranty

Repairs - What you need to know ⇒Page 30.

Warranty conditions ⇒Page 31.

11.1 Repairs - What you need to know

Repairs can be carried out by any workshop authorised to perform measures on Volkswagen industrial engines. In the case of minor tasks that can be carried out with the engine installed, drive to a workshop with the equipment. If this is not possible, request onsite assistance.

If major repairs are needed, remove and clean the engine and send it to the approved workshop.

The workshop is not obliged to remove and replace the industrial engine or to repair auxiliary equipment not supplied by Volkswagen AG.

As the range of equipment is extensive, some special replacement parts for industrial engines might not be in stock at all of the approved workshops. However, all of the replacement parts are held by the main Volkswagen industrial engine suppliers and are therefore available at short notice.

11.2 Warranty conditions

The purchaser is permitted to pass on this warranty provided he passes on the service schedule for Volkswagen industrial engines to his customers. The following conditions apply. In this case, the term of the warranty starts when the product is delivered to the purchaser's customer.

1 - The vendor warrants that the object sold is free of faults. "Free of faults" means state of the art technology as used in products comparable to the object sold on the day of delivery. The warranty begins on the day of delivery. It is valid for a period of 24 months but is limited to a maximum of 2000 operating hours. If the operating hours are reached within the first 12 months, the period of warranty cover is 12 months.

2 - The purchaser is entitled to have any faults rectified (corrective measures). The following terms apply:

- The purchaser can claim to have corrective measures carried out on a Volkswagen industrial engine by the vendor or in a workshop authorised by Volkswagen AG to service/repair the object sold. The purchaser shall report faults to the workshop of choice as soon as they are detected. The faults must be reported in writing, or the workshop of choice invited to report on the faults.
- Corrective measures must be carried out immediately in accordance with technical requirements by replacing or repairing the defective parts. The purchaser shall not be liable for expenses incurred from the corrective measures. This applies in particular to labour costs and material costs. Travel expenses and transport costs will not be reimbursed. Replaced parts become the property of the vendor. If, within the course of the corrective measures, additional service work is necessary as specified by Volkswagen AG, the vendor will be responsible for these charges, including the cost of the required materials and lubricants.
- Parts fitted during the corrective measures are covered by the warranty granted to the object sold on the basis of the sales contract for the stated period.

Note

If the fault cannot be rectified or the purchaser considers further corrective measures to be unviable, the purchaser can demand an annulment (cancellation of contract) or a price reduction (lower sale price) from the vendor instead of the corrective measures.

3 - Claims are not accepted for faults and damage in the following circumstances:

- if the purchaser has not reported a fault or not had it recorded, or
- if the purchaser has not provided an opportunity for corrective measures immediately on being requested to do so, or
- if the object sold has been treated negligently or subjected to excessive load, e.g. racing competitions, or
- if the object sold had previously been inadequately repaired, serviced or maintained in a workshop that was not approved by Volkswagen AG, and if this should have been apparent to the purchaser or
- if parts which have not been authorised by Volkswagen AG are installed in the object sold or if the object sold has been modified in a manner not approved by Volkswagen AG, or
- if the purchaser has failed to observe instructions on operation, service and maintenance of the object sold.

4 - Natural wear and tear is excluded from the warranty.

5 - If the workshop approached by the vendor to rectify the faults causes a delay in the corrective measures, the purchaser has the right to withhold a reasonable portion of an outstanding payment on the object sold until the corrective measures are complete.

6 - Where the assured repair standards are not reached, the entitlement to compensation on the grounds of non-fulfilment is not affected.

7 - The warranty claims detailed above are subject to a period of limitation which ends with the warranty period defined in point 1. In the case of faults for which claims for damage are asserted during the warranty period, but which are not remedied before the warranty period expires, the warranty is extended for this fault until this fault has been remedied. The limitation period is interrupted for this fault. In cases relating to the second sentence, the period of limitation shall end 3 months after the workshop of choice declares the fault rectified or that no fault exists.

12 Glossary

These explanations refer to the „maintenance instructions“ only.
They are in no way intended to cover all areas of engineering.

Term	Explanation
Common rail „CR“	A common high pressure injection line or „rail“ that supplies fuel to all of the cylinders of the respective cylinder bank..
DIN	Deutsches Institut für Normung e. V. (German Institute for Standardisation)
Part no.	Replacement part number or spare part number
EN	European Standard
ETKA	Electronic parts catalogue that replaced the microfiche system

Term	Explanation
EOBD	European On-Board Diagnosis
MIL	„Malfunction Indicator Light“ - American designation for exhaust gas warning lamp K83
OBD	Onboard diagnosis (OBD) monitors all of the components that play a role in the exhaust gas emissions quality
OBD-II	American onboard diagnosis
Readiness code	8-digit binary code that indicates whether engine management has performed all exhaust-gas-relevant diagnoses
RON	„ “Stands for „Research Octane Number“, which is a measurement for the knock resistance of petrol
RPF	Particulate filter
SAE	Stands for „Society of Automotive Engineers“, which draws up proposals/guidelines on how legislation can be implemented, e.g. industrial standards
SRE	Intake manifold injection
VEP	Distributor-type injection pump
ULEV	Stands for "Ultra Low Emission Vehicle"